Bank Management System

Code and Branch Coverage Report
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Code Coverage

Line coverage is measured in C++ implementation sources; the interactive src/main.cpp entry point is excluded.

Source file	Covered	Total	Coverage
src/infra/AccountRepository.cpp	76	102	74.51%
src/infra/AuditLogger.cpp	20	20	100.00%
src/infra/Config.cpp	37	43	86.05%
src/infra/CredentialStore.cpp	56	63	88.89%
src/infra/CustomerAccountStore.cpp	190	218	87.16%
src/infra/DataIntegrityValidator.cpp	149	158	94.30%
src/infra/FileUtil.cpp	34	38	89.47%
src/infra/InputValidator.cpp	58	59	98.31%
src/infra/Journal.cpp	136	187	72.73%
src/infra/LoginService.cpp	27	27	100.00%
src/infra/PasswordHasher.cpp	78	81	96.30%
src/infra/TimeUtil.cpp	9	9	100.00%
src/infra/TransactionRepository.cpp	86	116	74.14%
src/scaffold_placeholder.cpp	0	1	0.00%
src/service/AccountClosureService.cpp	27	27	100.00%
src/service/AccountDetailsService.cpp	14	15	93.33%
src/service/AccountNumberGenerator.cpp	7	7	100.00%
src/service/AccountService.cpp	82	97	84.54%
src/service/AccountUnlockService.cpp	17	21	80.95%
src/service/AuditLogReviewService.cpp	42	45	93.33%
src/service/AuthService.cpp	17	19	89.47%
src/service/CustomerSearchService.cpp	56	60	93.33%
src/service/CustomerUpdateService.cpp	21	25	84.00%
src/service/RbacGuard.cpp	6	6	100.00%
src/service/SessionManager.cpp	25	25	100.00%

Source file	Covered	Total	Coverage
src/service/SystemStatusService.cpp	23	23	100.00%
src/service/TransactionService.cpp	326	362	90.06%
TOTAL	**1619**	**1854**	**87.32%**

Raw GCC Branch Outcomes

Raw gcov counts every compiler-reported branch outcome, including exception edges. Headers, tests, and src/main.cpp are excluded.

Source file	Covered	Total	Coverage
src/infra/AccountRepository.cpp	81	180	45.00%
src/infra/AuditLogger.cpp	35	58	60.34%
src/infra/Config.cpp	31	58	53.45%
src/infra/CredentialStore.cpp	59	106	55.66%
src/infra/CustomerAccountStore.cpp	182	318	57.23%
src/infra/DataIntegrityValidator.cpp	170	308	55.19%
src/infra/FileUtil.cpp	35	60	58.33%
src/infra/InputValidator.cpp	61	76	80.26%
src/infra/Journal.cpp	132	332	39.76%
src/infra/LoginService.cpp	42	84	50.00%
src/infra/PasswordHasher.cpp	43	66	65.15%
src/infra/TimeUtil.cpp	4	8	50.00%
src/infra/TransactionRepository.cpp	91	230	39.57%
src/scaffold_placeholder.cpp	0	0	0.00%
src/service/AccountClosureService.cpp	27	42	64.29%
src/service/AccountDetailsService.cpp	15	72	20.83%
src/service/AccountNumberGenerator.cpp	10	16	62.50%
src/service/AccountService.cpp	85	164	51.83%
src/service/AccountUnlockService.cpp	12	20	60.00%
src/service/AuditLogReviewService.cpp	38	64	59.38%
src/service/AuthService.cpp	15	26	57.69%
src/service/CustomerSearchService.cpp	34	51	66.67%
src/service/CustomerUpdateService.cpp	23	36	63.89%
src/service/RbacGuard.cpp	4	4	100.00%
src/service/SessionManager.cpp	6	8	75.00%

Source file	Covered	Total	Coverage
src/service/SystemStatusService.cpp	11	20	55.00%
src/service/TransactionService.cpp	384	738	52.03%
TOTAL	**1630**	**3145**	**51.83%**

Non-Throw Branch Outcomes

This supplemental metric excludes only outcomes explicitly marked as throw edges by gcov; it is not a replacement for raw branch coverage. Outcomes are counted as covered when taken at least once.

Source file	Covered	Total	Coverage
src/infra/AccountRepository.cpp	81	124	65.32%
src/infra/AuditLogger.cpp	35	37	94.59%
src/infra/Config.cpp	30	42	71.43%
src/infra/CredentialStore.cpp	59	75	78.67%
src/infra/CustomerAccountStore.cpp	182	222	81.98%
src/infra/DataIntegrityValidator.cpp	170	212	80.19%
src/infra/FileUtil.cpp	35	38	92.11%
src/infra/InputValidator.cpp	61	65	93.85%
src/infra/Journal.cpp	132	220	60.00%
src/infra/LoginService.cpp	42	58	72.41%
src/infra/PasswordHasher.cpp	43	54	79.63%
src/infra/TimeUtil.cpp	4	4	100.00%
src/infra/TransactionRepository.cpp	91	156	58.33%
src/scaffold_placeholder.cpp	0	0	0.00%
src/service/AccountClosureService.cpp	27	28	96.43%
src/service/AccountDetailsService.cpp	15	67	22.39%
src/service/AccountNumberGenerator.cpp	10	10	100.00%
src/service/AccountService.cpp	85	122	69.67%
src/service/AccountUnlockService.cpp	12	15	80.00%
src/service/AuditLogReviewService.cpp	38	51	74.51%
src/service/AuthService.cpp	15	22	68.18%
src/service/CustomerSearchService.cpp	34	38	89.47%
src/service/CustomerUpdateService.cpp	23	34	67.65%
src/service/RbacGuard.cpp	4	4	100.00%
src/service/SessionManager.cpp	6	6	100.00%

Source file	Covered	Total	Coverage
src/service/SystemStatusService.cpp	11	11	100.00%
src/service/TransactionService.cpp	384	488	78.69%
TOTAL	**1629**	**2203**	**73.94%**

Coverage was collected with GCC gcov after the instrumented CTest run. A missing or unmeasured source file is reported as an error instead of being silently omitted.